

MINDNOTES : A Mobile Platform to enable users to break stigma around mental health and connect with therapists

Prateek Chanda
Microsoft Research India
Bangalore, India
v-pratec@microsoft.com

Amogh Wagh
Microsoft Research India
Bangalore, India
amoghwagh@gmail.com

Jemimah Johnson
National Institute of Mental Health
and Neuro-sciences
India
jemimah2496@gmail.com

Swaraj Renghe
Microsoft Research India
Bangalore, India
t-srenghe@microsoft.com

Vageesh Chandramouli
Microsoft Research India
Bangalore, India
vachand@microsoft.com

George Mathews
Icarus Nova
Bangalore, India
george@icarus.co.in

Sapna Behar
Icarus Nova
Bangalore, India
sapna@icarusnova.com

Poornima Bhola
National Institute of Mental Health
and Neuro-sciences
India
poornimabhola@gmail.com

Girish Rao
National Institute of Mental Health
and Neuro-sciences
India
girishn@nimhans.ac.in

Paulomi Sudhir
National Institute of Mental Health
and Neuro-sciences
India
paulomi.sudhir@gmail.com

T. K. Srikanth
International Institute of Information
Technology, Bangalore
India

Amit Sharma
Microsoft Research India
Bangalore, India
amshar@microsoft.com

Seema Mehrotra
National Institute of Mental Health
and Neuro-sciences
India
seema@nimhans.ac.in

ABSTRACT

We describe a mobile app for helping people manage their mental health, keeping in mind the unique challenges offered by the Indian context. One of the biggest challenges in India is the stigma around mental health and consequently the concerns and misconceptions about reaching out for professional help. Therefore, the proposed app focuses on tackling the barriers that people experience when reaching out for help and provides a free-of-cost platform for interactions with mental health professionals to alleviate any concerns and obtain an informed view. All content is developed by clinical psychologists and rather than replacing professional help, the app provides an example of a technology intervention for mental health that supplements existing infrastructures of mental healthcare.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CSCW '21 Companion, October 23–27, 2021, Virtual Event, USA

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8479-7/21/10...\$15.00

<https://doi.org/10.1145/3462204.3482895>

KEYWORDS

Human centered computing, mental health, professional therapy, barriers

ACM Reference Format:

Prateek Chanda, Amogh Wagh, Jemimah Johnson, Swaraj Renghe, Vageesh Chandramouli, George Mathews, Sapna Behar, Poornima Bhola, Girish Rao, Paulomi Sudhir, T. K. Srikanth, Amit Sharma, and Seema Mehrotra. 2021. MINDNOTES : A Mobile Platform to enable users to break stigma around mental health and connect with therapists . In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '21 Companion)*, October 23–27, 2021, Virtual Event, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3462204.3482895>

1 INTRODUCTION

Every 1 in 4 people experience anxiety and depression at some point in their lives [11]. In India, nearly 15% of adults are in need of active professional help (e.g., psychotherapy, medication) for one or more mental health conditions [4]. Yet mental healthcare remains inaccessible for many, for two main reasons. Firstly, in low and middle income countries there is a shortage of mental health professionals. In India, for example, there are less than 1 mental health professionals per 100,000 people [11]. Secondly, people living with

mental health conditions often face several barriers to seek help. Due to stigma and lack of awareness [6, 13], people often presume that the challenges are unique to them or that it is a passing phase, and prefer self-reliance or reliance on informal sources of help even when these may be proving to be insufficient. Majority of apps in recent times provide therapy in a fully automated way or provide therapy via actual therapists. However, the underlying assumption includes that a person is already familiar about how therapy works. Hence these platforms fail to raise that awareness about therapy. Our proposed framework thus aims at it, by guiding users through self help modules to know/discover about their symptoms via algorithm evaluation and a specific section called *Breaking Barriers* to acquaint themselves with common barriers that people usually feel while seeking out professional help. In addition, we nudge users based on their responses and activities towards *Professional Connect* in the app, where they can connect with therapists via chat interface and audio/video consultation.

2 RELATED WORK

In recent years, many mobile apps have been proposed for helping people manage their mental health (see [1] for a review). Most of these apps focus on empowering an individual through self-help, often expecting users to go through the self-help programs entirely on their own. [15] discusses on the design process for building such self help tools for anxiety management through SAM-Selfhelp for Anxiety Management. [12] on the other hand presents design intervention primarily in conversation interface space through voice-bot for ADHD users. Others do provide the opportunity for consultation with a mental health (MH) professional, but such access is typically paywalled. In the Indian context, examples of such apps are Wysa [5] which provides a text-based conversational interface and Innerhour [7] that focuses on creating a personalized plan for each user based on their previous assessments.

A potential challenge with such mental health apps is that they do not communicate the limitations of self-help strategies [8]. While useful, self-help is a low-intensity intervention and needs to be complemented with professional help based on the severity of symptoms experienced by a person. Specifically, technology-based help-seeking interventions have the potential to reduce psychological barriers and improve accessibility to evidence-based mental health services [2, 3]. There has been a recent increase in efforts to develop technology-based interventions for providing access to mental healthcare [14], delivered through internet-based modules/websites, text messages, and mobile apps. The functions in these platforms range from symptom identification, self-care, supporting recovery to encouraging help-seeking attitudes and interventions aiming at prevention and treatment [9, 10]. Most of these, however, are developed in upper middle- and high-income countries like Australia and US and focus less on stigma, limiting the scope of application to countries like India. To the best of our knowledge, there is a severe dearth of initiatives in the Indian context that utilize technology for improving help-seeking for common mental health concerns. In this work, therefore, we focus on using technology to bridge this gap between self-help and professional help, and design a system that helps people make the transition when needed.

3 DESIGN

MindNotes recognizes that a change in inclination or attitude to seek professional help may happen through different pathways for different people. For some, helping them to work through known barriers may provide the impulse to reach out to a professional, for others it may be the taste of a professional interaction which could reassure them and give them confidence that it is okay to seek professional help. Yet others may have the need some form of recognition of their condition over a period of time which gives them the awareness that they are required to meet a professional.

The interactions within major sections of the app viz. *Self-discovery*, *Breaking Barriers*, and Self Help Modules like *Managing Worry*, *Negative Thought Traps*, *Managing Anger* are organized to cater to these different pathways. While Self Discovery deals with getting the users to know what they are going through, whether they are experiencing anxiety and/or depression, *Breaking barriers* deals with barriers a person might have while seeking out professional help or therapy. In addition, MindNotes also provides a suite of "Self Help" modules aimed at alleviating low grade mental health issues like mild anxiety through activities and games under "Managing Worry" or mild depression through writing out/detecting negative words through "Negative Thought Traps".

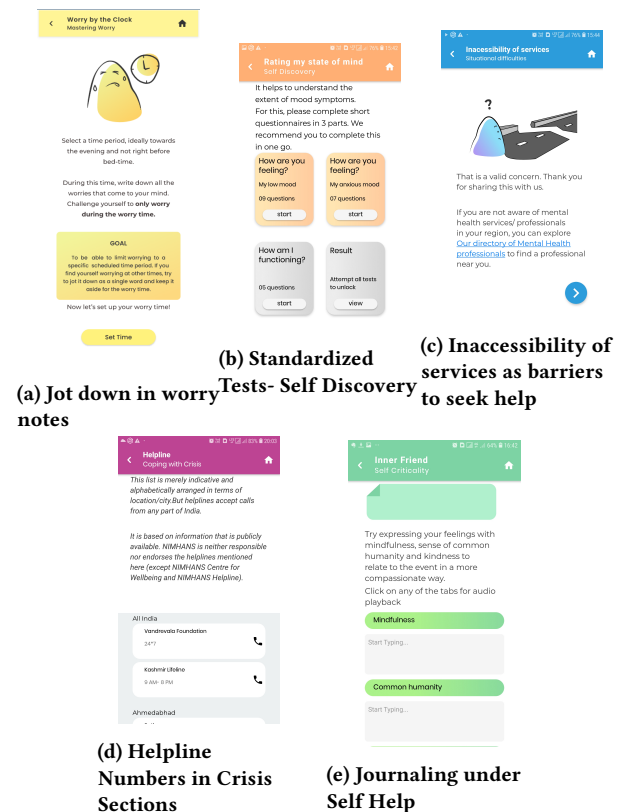


Figure 1: Screenshots from the MindNotes app illustrating the different sections of the app.

3.1 Breaking Barriers to seeking Professional Help

The goal of this section is to help people characterize and address their barriers to seeking professional help. Johnson et al. [6] list different barriers that prevent a person from reaching out, such as the reaction from friends or family, perception of being considered weak, or doubts about the usefulness of therapy. The app presents a set of these barriers to a user and asks them to select the ones applicable to them. Using a mix of text and video content aimed to speak to the user's psychological concerns, the app then presents information and interactive exercises that may be useful in helping the user reduce the barrier.

3.2 Professional Connect

This section allows a user to have interactions via text and video/audio with mental-health professionals to discuss their ongoing mental health or any questions about seeking help. These interactions focus on the concerns or misconceptions that a user may have about therapy but are not intended to provide therapy itself. The goal is not to conduct therapy over the app, but rather expose people to real therapists and help them get acquainted with the process of therapy. In this way, the goal of the app is not to be a substitute for mental healthcare infrastructure or mental health professionals, but rather encourage more people to be acquainted with the process of therapy or availing professional mental-health services.

3.2.1 Note Sharing Session. Inspired by recent rise in chat interface based platforms in the likes of Facebook and WhatsApp, we showcase a chat interface framework for therapists and users to connect over. In each such session, (Fig: 2a -2b) an user can send a text message as a note query to a therapist to which the therapist can respond to. Each such session requires cooperation on both sides to follow up on previous responses, after which it is closed. Unlike popular chat systems like WhatsApp, where only a new message arrival results in a notification popup, we also provide a reminder service (through email) as a nudge if users or therapists haven't followed up since their last message in a particular session. While users can share their queries or other info with a therapist, based on user's consent, the user's severity reports from standardized test scores and other self help activity logs can also be shared with the therapist, for them to better understand the case.

3.2.2 Audio/Video Connect. Similar to the note-sharing app, the audio-video sharing session lets the user interact with their therapist of choice only via audio or video Fig 2c. A unique feature of this framework is it provides asynchronous audio/video consultation, as compared to traditional e-consultation apps that provide face-to-face consultation. This is built keeping in mind and timing conflicts that may happen if a therapist is not available during the scheduled time. Also, in the Indian context, bandwidth issues often act as a barrier to real-time consultation. Using this feature, a user can send over a short 5 min audio/video clip detailing their query over to a therapist.

3.2.3 Notifications in Mail Inbox. Apart from the normal app interactions, the framework also provides a feature where users and therapists are notified of their session details in their mail inbox

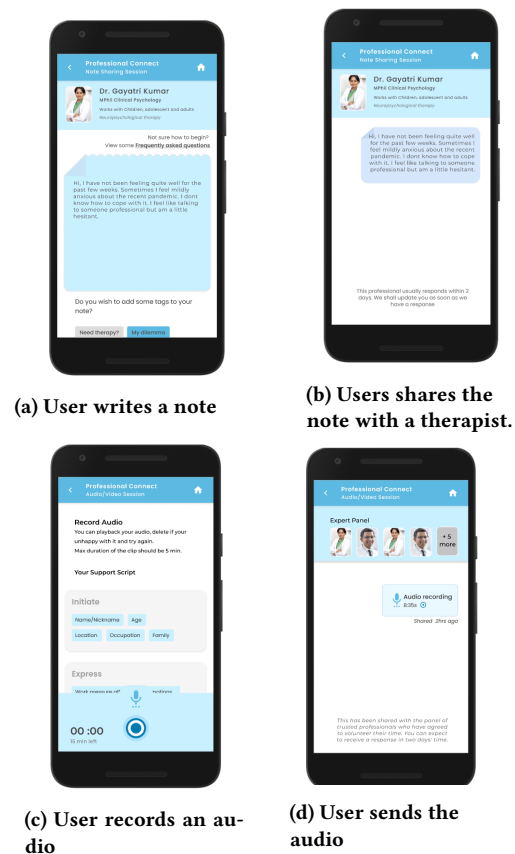


Figure 2: Note Sharing & Audio/Video Session - User-facing interface

as well. Since the message communication happens via Microsoft Office 365's Graph API Service, every user gets notified based on their email address stored on our server, of the recent message they received. Also, reminders to nudge therapists or clients to followup in their session is provided via background service to their corresponding emails. This feature is useful because therapists don't always have to use the app to get notified of the recent new message or a reminder about a session for which their followup is pending. They can simply get notified throughout the day through their mail inbox, without the need to check the app.

4 CONNECTING USERS WITH THERAPISTS

Each therapist can see a list of current new user requests 3c as well as a list of pending requests they have to follow up 3d. Hence it may cause certain delay to attend to some users, given too many incoming and pending requests. While getting connected to a therapist is crucial for users who may feel they need to go for therapy as soon as possible, it is also important that they connect to someone who is more approachable and has the relevant experience customized to the user's need as well as who can follow up well.

Users are free to visit various sections within the app including *Self Discovery*, *Self Help* modules and barriers to seeking help, even

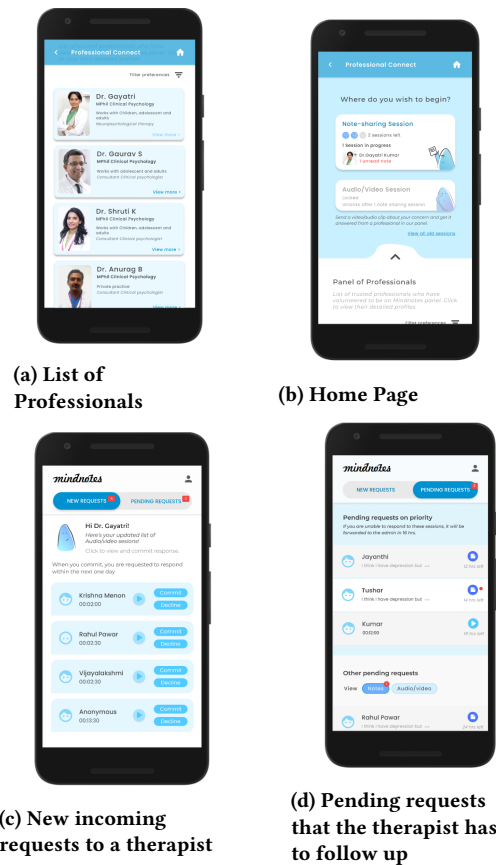


Figure 3: An Overview of the Professional Connect Section. 3a,3b show the user-facing interface and 3c,3d show the therapist's interface

before they embark on Professional Connect. Ranking of users is done based on the severity of symptoms (standardized test scores from PHQ9, GAD7) as well as the set of barriers they chose from *Breaking Barriers* section (collected from user responses and telemetry data of section visits).

Users with higher score are provided with a recommended therapist list whose individual professional expertise has a higher match with the symptoms they are facing, along with a high follow up rate. For a particular therapist, followup rate here refers to how many percentage of requests a therapist was able to follow up within time before which it is handed over to volunteers.

5 VOLUNTEER HANDLING THERAPIST'S PENDING REQUESTS

In case a therapist is unable to follow up with a user within 2 days under pending request (Fig.3d), the request is handed over to a volunteer who is available 24x7 to respond to such requests on behalf of the therapist. Volunteers are trained professionals and have broad range of professional expertise. This helps users get proper feedback and their queries resolved as soon as possible even when the therapist may not be available. This is a crucial feature

for someone in crisis or going through a tough phase for whom getting help and guidance is of the utmost importance.

6 TECHNICAL OVERVIEW & PRIVACY

The app is built using the React Native framework that allows the code base to work on Android and iOS devices. The front-end of the app is based on an open-source framework *Sahay*¹, developed for building content-heavy apps. This toolkit provides for easy localization and reuse of same UI components across various app sections. Given that the target audience in India have intermittent internet access, the app is designed with most of its content locally bundled such that it can work without any internet connection after it has been downloaded.

For the note sharing and audio/video section under Professional Connect, the message service was built using Microsoft Graph API with secure Microsoft Exchange service. Responses and telemetry from the users are treated as anonymized health records and are encrypted and backed up to a secure service.

7 DEMONSTRATION SETUP

We will be showcasing a live app demo and offer participants the opportunity to download the app themselves, or simply follow the demo on the presenter's phone/screen.

ACKNOWLEDGMENTS

We thank the research team involved at NIMHANS-IIITB for their insightful comments on the paper. We also thank Shilpa Bhat for helping us with self-help sections development. Lastly, we thank people at ICARUS - Nikita Arora, Varshapriya R who were involved in the initial design discussions for the app.

REFERENCES

- [1] David Bakker, Nikolaos Kazantzis, Debra Rickwood, and Nikki Rickard. 2016. Mental health smartphone apps: review and evidence-based recommendations for future developments. *JMIR mental health* 3, 1 (2016), e7.
- [2] D Ben-Zeev. 2014. Technology-based interventions for psychiatric illnesses: improving care, one patient at a time. *Epidemiology and psychiatric sciences* 23, 4 (2014), 317–321.
- [3] Bonnie A Clough, Sonia M Nazareth, and Leanne M Casey. 2020. Making the grade: a pilot investigation of an e-intervention to increase mental health literacy and help-seeking intentions among international university students. *British Journal of Guidance & Counselling* 48, 3 (2020), 347–359.
- [4] G Gururaj, Mathew Varghese, V Benegal, G NIMHANS Rao, K Pathak, LK Singh, and R Misra. 2016. National mental health survey of India, 2015–16: Summary. *Bangalore: National Institute of Mental Health and Neurosciences* (2016).
- [5] Becky Inkster, Shubhankar Sarda, and Vinod Subramanian. 2018. An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: real-world data evaluation mixed-methods study. *JMIR mHealth and uHealth* 6, 11 (2018), e12106.
- [6] Jemimah A Johnson, Janhavi Devdutt, Seema Mehrotra, Poornima Bhola, Paulomi Sudhir, and Amit Sharma. 2020. Barriers to Professional Help-seeking for Distress and Potential Utility of a Mental Health App Components: Stakeholder Perspectives. *Cureus* 12, 2 (2020).
- [7] Amit Malik, Alisha Kalidhar, and Cécile Hanon. 2020. InnerHour: Solving the Treatment Gap in Mental Health. Using technology to disrupt the mental health landscape. *L'information psychiatrique* 96, 5 (2020), 348–354.
- [8] Seema Mehrotra, Satish Kumar, Paulomi Sudhir, Girish N Rao, Jagadisha Thirthalli, and Aditi Gandotra. 2017. Unguided mental health self-help apps: reflections on challenges through a clinician's lens. *Indian Journal of Psychological Medicine* 39, 5 (2017), 707.
- [9] Seema Mehrotra and Ravikesh Tripathi. 2018. Recent developments in the use of smartphone interventions for mental health. *Current opinion in psychiatry* 31, 5 (2018), 379–388.

¹github.com/microsoft/sahay-app-toolkit

- [10] Miranda Olff. 2015. Mobile mental health: a challenging research agenda. *European journal of psychotraumatology* 6, 1 (2015), 27882.
- [11] World Health Organization. 2001. *The World Health Report 2001: Mental health: new understanding, new hope*. World Health Organization.
- [12] Do Eun Park, Yee-Jin Shin, EunAh Park, In Ae Choi, Woo Yeon Song, and Jinwoo Kim. 2020. Designing a Voice-Bot to Promote Better Mental Health: UX Design for Digital Therapeutics on ADHD Patients. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*.
- [13] Sachin R Pendse, Naveena Karusala, Divya Siddarth, Pattie Gonsalves, Seema Mehrotra, John A Naslund, Mamta Sood, Neha Kumar, and Amit Sharma. 2019. Mental health in the global south: challenges and opportunities in HCI for development. In *Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies*. 22–36.
- [14] Matthew Price, Erica K Yuen, Elizabeth M Goetter, James D Herbert, Evan M Forman, Ron Acierno, and Kenneth J Ruggiero. 2014. mHealth: a mechanism to deliver more accessible, more effective mental health care. *Clinical psychology & psychotherapy* 21, 5 (2014), 427–436.
- [15] Phil Topham, Praminda Caleb-Solly, Paul Matthews, Andy Farmer, and Chris Mash. 2015. Mental health app design: A journey from concept to completion. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct*. 582–591.